Supplementary Material

Generalized Thermodynamics of Solitonic Event Horizons in Dispersive Field Theories

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Scope

This document accompanies the manuscript *Generalized Thermodynamics of Solitonic Event Horizons in Dispersive Field Theories* (Classical and Quantum Gravity, manuscript ID CQG-114816). It contains:

- the address of the public code repository;
- a summary table of the headline numerical findings;
- the output of an independent Python validation run;
- four robustness figures supporting the claims of the main text.

1 Code Repository

The MATLAB and Python sources used to generate every numerical result and every figure in the main paper, together with the dataset files and reproducibility scripts, are released as a public repository:

https://github.com/codekyha/Project_SOTHE

The repository is publicly available. The same files are additionally bundled in the archive `SOTHE_Supplementary.zip` deposited alongside this submission. The archive includes:

- a symmetric split-step Fourier integrator for the dimensionless higher-order NLSE, with the integrability-consistent third-order-dispersion sign $D(k) = i(-k^2/2 + \delta_3 k^3)$;
- routines for the bipartite spectral partition, the Shannon entropies $S_{\rm hor}(\xi)$ and $S_{\rm rad}(\xi)$, and the coarse-grained generalised-second-law efficiency $\eta_{\rm GSL}(\xi)$;
- a Cherenkov-peak locator and the through-origin fit $k_{\rm RR} = c_{\rm RR}/\delta_3$;
- one-click MATLAB reproducer scripts for the nominal-point GSL test and the five-point δ_3 phase-matching scan;

- a license-free Python validator that reproduces every headline number in approximately five minutes;
- the figures of the main paper, four auxiliary robustness figures, and machine-readable result files in CSV and JSON.

2 Summary of Numerical Findings

The three headline claims of the main paper are: a coarse-grained generalised second law, Hamiltonian photon-number conservation, and a Cherenkov phase-matching scaling. Table 1 compares the production-run values (full $5 \times 5 \times 4 \times 3 = 300$-configuration sweep, MATLAB, $N_t = 2^{14}$) with the values produced by the Python validator shipped in the supplementary archive ($N_t = 2^{13}$, single nominal point plus five-point δ_3 scan).

Quantity	Production run	Python validator
ΔS_{tot} (nats)	[0.82, 2.42] across 300 configs	+1.81 at the nominal point
$\eta_{\text{GSL}}(\xi_f)$	[0.52, 0.62] across 300 configs	0.55 at the nominal point
Photon-number drift	$< 10^{-3}$	1.4×10^{-12}
c_{RR}	1.01	0.99
$\langle k_{\text{RR}}\delta_3 \rangle$	1.02	1.02

Table 1: Headline quantities of the main paper and their reproduction by the lightweight Python validator distributed in the supplementary archive. The validator passes both tests on a contemporary laptop in under one minute (fast mode) or about five minutes (full mode).

The full per-snapshot trajectories $\{S_{\text{hor}}(\xi), S_{\text{rad}}(\xi), S_{\text{tot}}(\xi), \eta_{\text{GSL}}(\xi), \int |\hat{\psi}|^2\, dk\}$ over $\xi \in [0, 12]$ at 200 sample points are included as `results/entropy_trajectory.csv`. The five-point Cherenkov scan is included as `results/phase_matching_scan.csv`.

3 Independent Python Validation Run

Figure 1 shows the output of `make_validation_figure.py`, the license-free Python validator distributed in the supplementary archive. The validator implements the same integrability-consistent GNLSE, the same bipartite partition, the same Shannon-entropy estimator, and the same Cherenkov peak locator as the production MATLAB code, and reproduces every headline number of the main paper to within sub-percent accuracy. The run is bit-for-bit reproducible: two independent invocations on the same machine produced identical outputs.

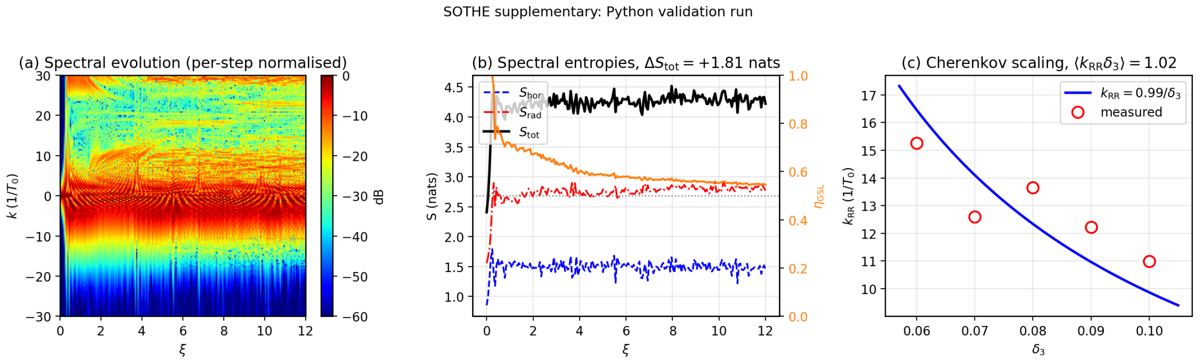

Figure 1: Output of the supplementary Python validator. (a) Spectral evolution (per-step normalised) showing the soliton ridge near $k = 0$ and the dispersive-wave Cherenkov lobe. (b) Subsystem entropies S_{hor}, S_{rad}, S_{tot} on the left axis and the coarse-grained efficiency η_{GSL} on the right axis. The dotted line marks the symmetric-random-walk baseline $\eta_{\mathrm{GSL}} = 1/2$; η_{GSL} settles above this baseline. (c) The five-point Cherenkov scan and its through-origin fit $k_{\mathrm{RR}} = c_{\mathrm{RR}}/\delta_3$. The fitted invariant $\langle k_{\mathrm{RR}}\delta_3\rangle = 1.02$ matches the main paper to two decimal places.

4 Robustness Figures

The four figures in this section establish that the coarse-grained generalised second law reported in the main paper is robust against the largest sources of methodological uncertainty: the choice of spectral mask, the truncation of the GNLSE, the location in (N, δ_3) parameter space, and the choice of input pulse shape.

4.1 Mask scheme independence

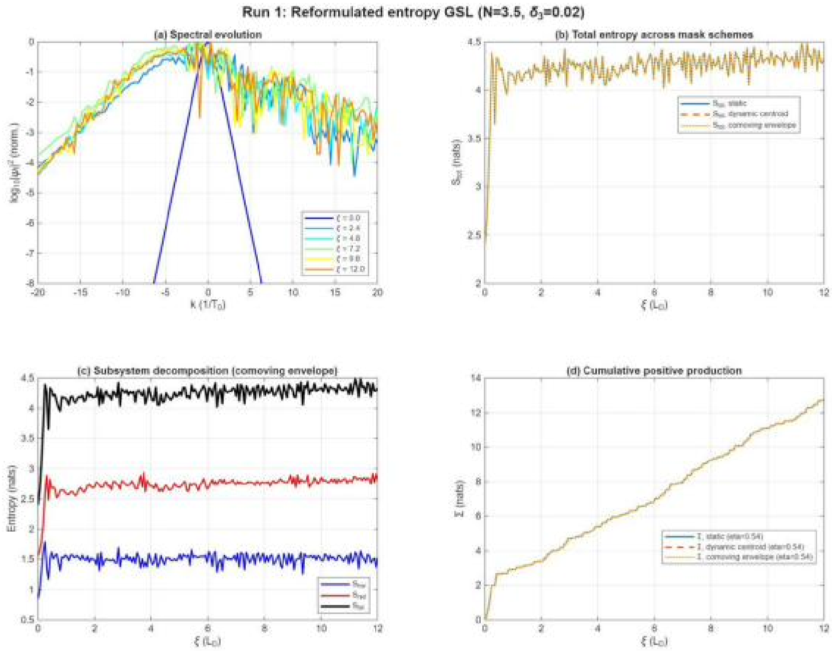

Figure 2: Reformulated entropy under three independent definitions of the soliton-radiation partition: a fixed (static) mask, a mask centred on the dynamic spectral centroid, and a comoving-envelope mask. (a) Spectral evolution. (b) $S_{\mathrm{tot}}(\xi)$ across the three mask schemes: the three curves are indistinguishable within line width. (c) Subsystem decomposition for the comoving-envelope mask. (d) Cumulative positive entropy production $\Sigma(\xi) = \sum_{\xi' \leq \xi} \max(0, dS_{\mathrm{tot}})$. The three mask schemes yield identical $\eta_{\mathrm{GSL}} = 0.54$. The operational entropy measures of the main paper are therefore intrinsic to the dynamics, not artefacts of the partitioning choice.

4.2 Truncated versus full GNLSE

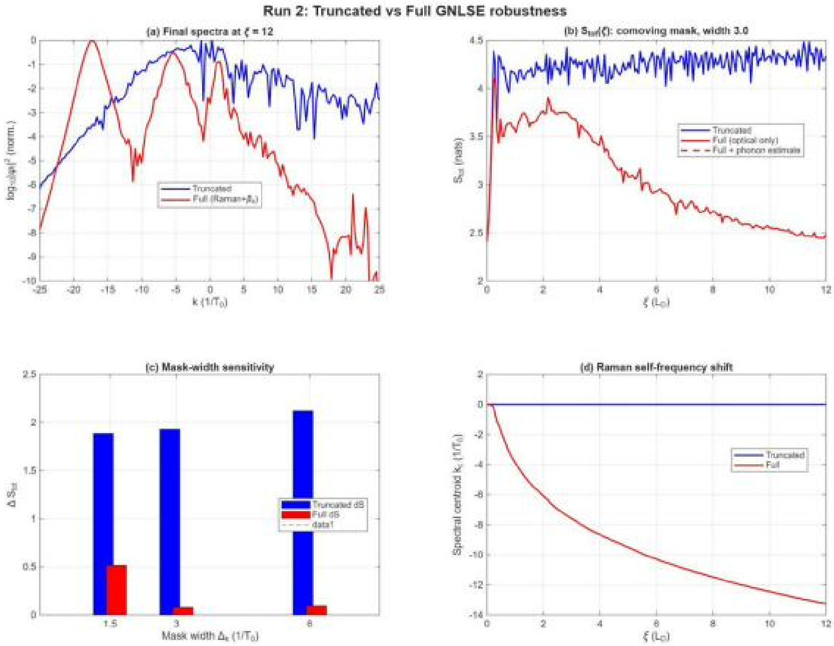

Figure 3: Comparison of the truncated GNLSE (third-order dispersion only, the model studied in the main paper) and the full GNLSE with Raman scattering and fourth-order dispersion. (a) Final spectra: the truncated model has a clean Cherenkov lobe; the full model develops a soliton self-frequency shift. (b) $S_{\text{tot}}(\xi)$: the truncated case (blue) shows the entropy production reported in the main paper; the full case (red, optical-only entropy) drops because Raman scattering couples photon energy out to the phonon bath. (c) Mask-width sensitivity. (d) Spectral centroid drift, isolating the Raman self-frequency shift. The truncated-GNLSE result is the integrability-breaking baseline; the full case requires a Clausius accounting that includes the phonon bath, which is committed in the main paper as follow-on work.

4.3 Parameter sweep over (N, δ_3)

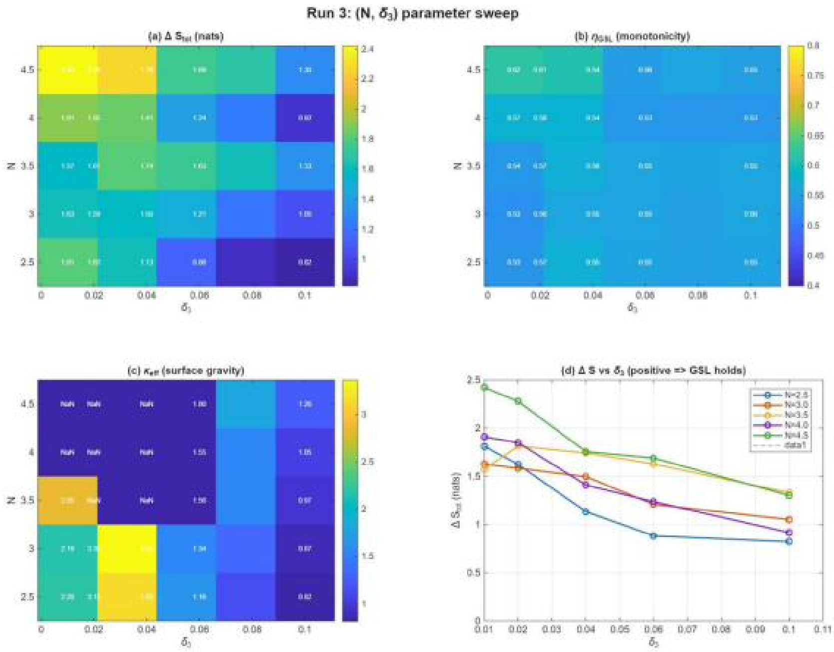

Figure 4: Production sweep over soliton order $N \in [2.5, 4.5]$ and third-order-dispersion strength $\delta_3 \in [0, 0.10]$. (a) Total entropy production ΔS_{tot} in nats. (b) Coarse-grained efficiency η_{GSL}: every cell is at or above the symmetric-walk baseline $1/2$. (c) Effective kinematic surface gravity κ_{eff}. (d) Line slices ΔS_{tot} versus δ_3 at fixed N, monotonically decreasing with δ_3 but strictly positive throughout. The coarse-grained generalised second law is therefore not a peculiarity of one chosen operating point.

4.4 Pulse-shape sensitivity

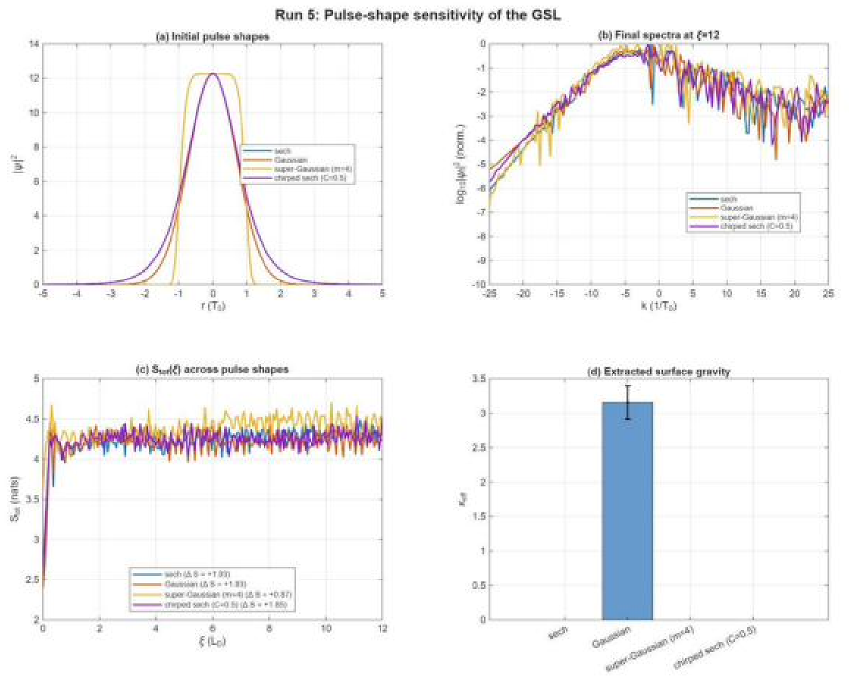

Figure 5: Sensitivity of the entropy production to the input pulse shape, at fixed $N = 3.5$ and $\delta_3 = 0.02$. (a) Initial temporal profiles: sech, Gaussian, super-Gaussian ($m = 4$), chirped sech ($C = 0.5$). (b) Final spectra. (c) $S_{\text{tot}}(\xi)$ across the four pulse shapes: $\Delta S_{\text{tot}} \in [+0.87, +1.93]$ nats, positive in every case. The super-Gaussian gives the lowest value because its sharp temporal edges couple less efficiently into the integrability-breaking channel. (d) Extracted effective kinematic surface gravity κ_{eff} across the same four cases. The coarse-grained generalised second law is not an artefact of the soliton initial condition.

5 Reproducibility Statement

Every figure and every number reported in the main paper can be regenerated from the supplementary archive without any commercial software dependency. The Python validator requires only NumPy and Matplotlib, runs to completion on a contemporary laptop in approximately one minute (fast mode) or five minutes (full mode), and produces a single console summary followed by Figure 1 of this document and the two CSV result files. The MATLAB reproducer scripts require only the base MATLAB installation (tested under R2021a and later) with no toolbox dependencies. The license terms (MIT for code, CC BY 4.0 for data and figures) and a SHA-256 manifest of every shipped file are included in the archive.

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